ModelIt!

Student Activity Pack

How Do Ecosystems Recover from Catastrophe?

Modeling Ecological Succession, Resilience Thresholds, and Regime Shifts

Grade 12L3 | NGSS HS-LS2-6 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **If a wildfire completely destroys a forest, what do you think happens first in the recovery process? Describe the sequence of changes you would expect.**

2. **What is the difference between an ecosystem recovering to its original state and transforming into a completely different type of ecosystem?**

3. **Why might some ecosystems recover quickly from disturbance while others take centuries or never fully recover?**

4. **How might climate change affect an ecosystem's ability to recover from a catastrophic disturbance like a volcanic eruption or massive wildfire?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Disturbance Severity
- Seed Bank and Propagule Availability
- Climate Suitability
- Pioneer Species Establishment
- Soil Development Rate
- Biodiversity Accumulation
- Nutrient Cycling Efficiency
- Trophic Web Complexity
- Invasive Species Pressure
- Recovery Completeness Index

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Post-Fire Recovery

Disturbance: Moderate | Soil: Intact | Seed Bank: Rich | Climate: Stable

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Primary Succession

Disturbance: Complete sterilization | Soil: None | Seed Bank: None | Climate: Stable

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Climate-Shifted Recovery

Disturbance: High | Soil: Degraded | Seed Bank: Moderate | Climate: 2 degrees C warmer, 20% drier

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Mycorrhizal Network Recovery — The restoration of underground fungal networks that connect plant roots, facilitate nutrient transfer between trees, and provide chemical communication pathways — mature forests have extensive mycorrhizal networks that take decades to re-establish and without which tree seedlings grow 50-80% more slowly

Keystone Species Reintroduction — The deliberate return of species whose ecological impact is disproportionately large relative to their abundance — wolves, beavers, prairie dogs, and certain trees function as ecosystem engineers whose absence fundamentally alters succession pathways and community structure

Legacy Effect Strength — The degree to which remnants of the pre-disturbance ecosystem (surviving trees, buried seeds, root networks, soil microbiome) persist and influence recovery — stronger legacy effects accelerate succession and increase the probability of recovering toward the original ecosystem state

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

After a volcanic eruption sterilizes the landscape, a wildfire scorches a forest, or an oil spill devastates a coastline, the ecosystem does not simply regrow — it rebuilds through a complex sequence of pioneer species, competitive replacements, and food web reconstruction that can take decades or centuries. What determines whether an ecosystem recovers to its original state, shifts to a completely different ecosystem, or collapses entirely? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate why soil condition is the critical bottleneck that determines whether recovery takes decades or centuries?**

2. **What does the regime shift scenario reveal about the limits of ecosystem resilience, and how does climate change lower those limits?**

3. **Why does the order in which species colonize matter for the long-term recovery trajectory, and how can invasive species permanently alter that trajectory?**

4. **How would adding mycorrhizal networks or keystone species to your model change the predicted Recovery Completeness Index and timeline?**

5. **What are the ethical implications of 'assisted migration' — deliberately planting species adapted to future climate conditions rather than trying to recreate historical ecosystems?**

8 STEM Challenge: Design an Ecosystem Restoration Plan

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YOUR ENGINEERING MISSION

Design a comprehensive restoration plan for an ecosystem devastated by catastrophic disturbance, specifying the sequence of interventions needed to guide recovery toward the desired state while accounting for climate change, invasive species, and resilience thresholds.

THE SCENARIO

A severe wildfire has burned 50,000 acres of mixed-conifer forest in the Sierra Nevada. Soil is severely damaged in 30% of the area, moderate in 50%, and intact in 20%. Climate projections indicate the region will be 1.5 degrees C warmer and 15% drier by 2050, which may not support the original conifer forest. Invasive cheatgrass is already colonizing burned areas. Your restoration ecology team must design a plan that maximizes Recovery Completeness Index while adapting to projected climate conditions — which may mean planting species adapted to future conditions rather than trying to recreate the past.

Engineering Design Thinking

1. WHAT IS THE TARGET ECOSYSTEM FOR YOUR RESTORATION PLAN AND HOW DOES IT ACCOUNT FOR PROJECTED CLIMATE SHIFTS?

2. WHAT SPECIFIC INTERVENTIONS DOES YOUR PLAN INCLUDE IN YEARS 1, 5, 10, AND 25, AND WHAT IS THE SCIENTIFIC RATIONALE FOR THE SEQUENCE?

3. HOW DOES YOUR PLAN ADDRESS THE 30% OF SEVERELY DAMAGED SOIL WHERE NATURAL RECOVERY WILL BE EXTREMELY SLOW?

4. WHAT MONITORING METRICS WILL YOU TRACK TO DETERMINE IF THE ECOSYSTEM IS ON A RECOVERY TRAJECTORY OR HEADING TOWARD A REGIME SHIFT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Restoration ecologists who design and manage ecosystem recovery projects earn $55,000-$100,000/year at conservation organizations and government agencies. Wildlife biologists who study species recovery after disturbance earn $50,000-$90,000/year. Fire ecologists who study how fire shapes ecosystems and manage prescribed burns earn $55,000-$95,000/year. Conservation geneticists who use DNA analysis to guide restoration plant and animal sourcing earn $60,000-$110,000/year.